

# USER\_GUIDE

## VisionEngine User Guide

### Installation

```
go get digital.vasic.visionengine
```

### Configuration

Set environment variables in `.env` or export them:

```
export HELIX_VISION_PROVIDER=auto
export OPENAI_API_KEY=sk-...
export ANTHROPIC_API_KEY=sk-ant-...
```

See `.env.example` for all configuration options.

### Using NavigationGraph

```
import (
    "digital.vasic.visionengine/pkg/analyzer"
    "digital.vasic.visionengine/pkg/graph"
)

// Create graph
g := graph.NewNavigationGraph()

// Add screens
g.AddScreen(analyzer.ScreenIdentity{ID: "home", Name: "Home
    Screen"})
g.AddScreen(analyzer.ScreenIdentity{ID: "settings", Name:
    "Settings"})

// Add transitions
g.AddTransition("home", "settings", analyzer.Action{Type:
    "click", Target: "gear"})

// Navigate
```

```
g.SetCurrent("home")
path, err := g.PathTo("settings")

// Export
dot := graph.ExportDOT(g)
jsonStr, _ := graph.ExportJSON(g)
mermaid := graph.ExportMermaid(g)
```

## Using LLM Vision Providers

```
import "digital.vasic.visionengine/pkg/llmvision"

provider, _ :=
    llmvision.NewOpenAIProvider(llmvision.ProviderConfig{
        APIKey: os.Getenv("OPENAI_API_KEY"),
    })

result, err := provider.AnalyzeImage(ctx, screenshotBytes, "What
do you see?")
```

## Fallback Chain

```
openai, _ := llmvision.NewOpenAIProvider(openaiConfig)
anthropic, _ := llmvision.NewAnthropicProvider(anthropicConfig)
fallback, _ := llmvision.NewFallbackProvider(openai, anthropic)

// Automatically falls back to Anthropic if OpenAI fails
result, err := fallback.AnalyzeImage(ctx, img, "Describe this
screen")
```